

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Tube cutter, Part Number 3376579

Additional Service Items

- No additional service items required.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

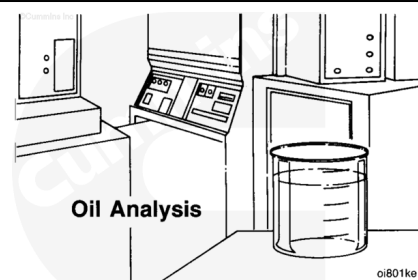
To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

- Disconnect the batteries. Reference the equipment manufacturer service information.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/40/40-007-037.html)

Inspect

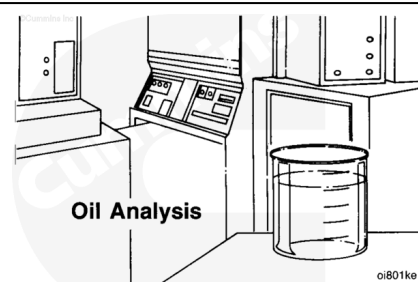
An analysis of used oil can help diagnose internal engine damage and determine if it was caused by one of the following:

- Intake air filter malfunction
- Coolant leaks
- Oil diluted with fuel
- Metal particles causing wear.



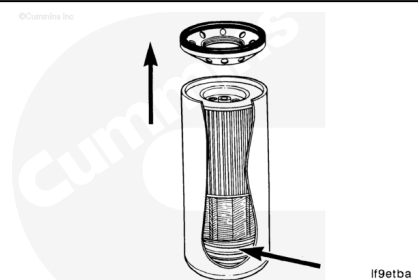
For additional oil analysis information, see Fluids for Cummins® Products Service Manual, Bulletin 5411406.

Note : Do not disassemble an engine for repair based solely on the results of an oil analysis. Inspect the oil filters. If an oil filter shows evidence of internal engine damage, find the source of the problem and repair the damage. Reference the appropriate procedure(s) based on the following oil filter inspection.



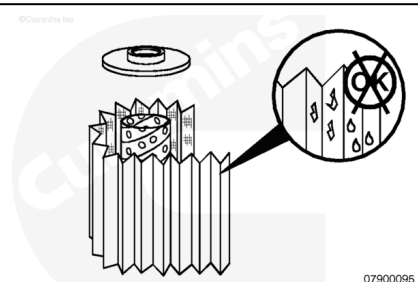
⚠ WARNING ⚠

Restrain the full flow lubricating oil filter and use care when cutting open the upper section of the combination filter. The filter element spring is under compression and can cause personal injury.



Use tube cutter, Part Number 3376579, to open the upper section of the bypass full-flow oil filter.

Inspect the filter element for evidence of moisture or metal particles.



Metal	Possible Source
Copper	Bearings and bushings
Chromium	Piston rings
Iron	Cylinder liners
Lead	Bearing overlay material
Aluminum	Piston wear or scuffing

Clean and Inspect for Reuse

CAUTION

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

For all other internal engine malfunctions which introduce debris into the lubricating oil system, complete the following:

- Replace lubricating oil filter (spin-on). Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-013-tr.html>)
- Replace lubricating oil cooler element. Refer to Procedure 007-003 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-003-tr.html>)
- Keep removed parts clean.
- Do **not** use cleaning agents other than contact cleaner.
- Clean lubricating oil cooler housing. Refer to Procedure 007-003 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-003-tr.html>)
- Clean the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html>)
- Remove and inspect the Lubricating oil pump for reuse. Refer to Procedure 007-031 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-031-tr.html>)
- Inspect connecting rod bearings. Refer to Procedure 001-005 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-005.html>)
- Inspect main bearings. Refer to Procedure 001-006 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-006.html>)

Finishing Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install lubricating oil pan and other components removed to access main bearings. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html>)
- Install lubricating oil filter. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-013-tr.html>)
- Prime and fill the engine with new lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-037.html>)

- Operate the engine to normal operating temperature and check for leaks and proper operation.
- Connect the batteries. Reference the equipment manufacturer service information.

Last Modified: 09-Feb-2023
